

NAME

codata – library for fundamental physical constants

LIBRARY

Codata library (*libcodata*, *-lcodata*)

SYNOPSIS

Fortran **use codata**

C **#include "codata.h"**

Python **import pycodata**

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants published by CODATA. It includes the 2022, 2018, 2014 and 2010 recommended values, allowing applications to reproduce calculations based on specific CODATA releases.

The constants are implemented as a derived type that carries: *the name, the value, the uncertainty, and the unit*. All the constants are provided as **double precision reals**. Long constant names can be aliased with shorter names.

It provides:

- CODATA 2022 constants
- CODATA 2018 constants
- CODATA 2014 constants
- CODATA 2010 constants
- Native Fortran API
- C interoperability API
- Python bindings

DOCUMENTATION

The API is written as POSIX-style man pages which are distributed with the binaries.

HTML <https://milanskocic.github.io/codata/index.html>

PDF <https://milanskocic.github.io/codata/man/codata-bookman.pdf>

NOTES

The codata constants from the 2022 adjustment were integrated into the Fortran standard library (stdlib). This library is complementary to the constants defined in the stdlib by providing older values for the constants. See <https://github.com/fortran-lang/stdlib/releases/tag/v0.7.0>. If your application only requires the latest CODATA values, the Fortran standard library may be sufficient. Use codata when you need access to historical CODATA releases or compatibility with older scientific results.

To use codata within your fpm(1) (<https://github.com/fortran-lang/fpm>) project, add the following lines to your file:

```
[dependencies]
codata = { git="https://github.com/MilanSkocic/codata.git" }
```

If you only need sources for the CODATA constants that can be integrated directly into your project, you may be interested in the following repository: https://github.com/vmagnin/fundamental_constants.

LICENSE

MIT

SEE ALSO

codata(7), codata(3), codata_cli(1), codata_version(3), codata_constant_type(3)